



Statistics of Financial Markets

Chapter 2: Statistical Distributions for Financial Markets



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Learning Objectives

By the end of this chapter, you will be able to:

1. **Explain** why the choice of distribution affects financial risk estimation
2. **Compute** log-returns and interpret their statistical properties
3. **Identify** stylized facts of returns (heavy tails, skewness, clustering)
4. **Compare** Normal, Student- t , stable, and EVT distributions
5. **Apply** Value-at-Risk and Expected Shortfall under different distributional assumptions
6. **Evaluate** GARCH models with non-Gaussian innovations

Contents

1. Introduction
2. From Prices to Log-Returns
3. The Normal Distribution
4. Stylized Facts
5. The Student- t Distribution
6. Asymmetric Distributions
7. Stable Distributions
8. Extreme Value Theory
9. Distribution Comparison
10. Risk Management
11. Modern Extensions

Why Does the Distribution Choice Matter?

- If we assume the wrong distribution, we **underestimate** the probability of extreme losses
- **Example:** October 19, 1987 — the S&P 500 fell 20.5% in a single day
 - ▶ Under the Normal assumption: a $\sim 25\sigma$ event with probability $\approx 10^{-135}$
 - ▶ Yet it happened — the Normal distribution is inadequate for tail events
- Banks and regulators need to quantify “how bad can it get?”
- This requires **accurate distributional models**

Key Insight

- The choice of probability distribution directly determines how much capital a bank must hold against potential losses

The Distribution Choice Problem

- **Preview:** In Section 2.10 we will formally define **Value-at-Risk (VaR)**, a key risk measure that depends entirely on the distributional assumption
- The candidates we will study:
 - ▶ **Normal** — the classical benchmark (thin tails)
 - ▶ **Student- t** — heavier tails, finite higher moments
 - ▶ **Stable** — heaviest tails, infinite variance possible
 - ▶ **EVT** — models only the tails, agnostic about the center
- Each model implies a different answer to “how bad can it get?”

Model	Tail Type	Kurtosis	Parameters
Normal	Exponential	3	2
Student- t	Power law	> 3	3
Stable	Power law	∞	4
GPD (EVT)	Flexible	—	2

Chapter Roadmap

Foundations

1. From prices to log-returns
2. Normal distribution (benchmark)
3. Stylized facts of financial returns
4. Student- t distribution
5. Asymmetric distributions

Advanced Models & Applications

6. Stable distributions
7. Extreme Value Theory
8. Distribution comparison
9. **Risk management (VaR, ES)**
10. Modern extensions (GARCH)

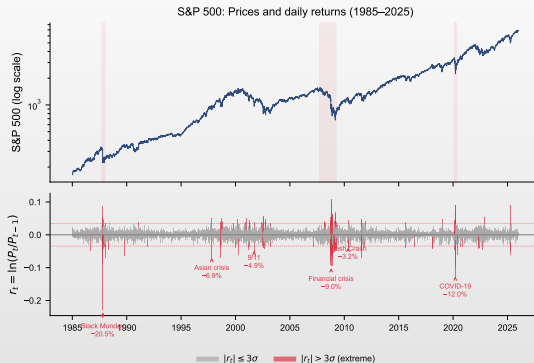
Distribution Choice and Derivative Pricing

- ▣ Black–Scholes assumes log-normality $\Rightarrow$ constant implied volatility
- ▣ Empirical deviations produce the **volatility smile/skew**
- ▣ Portfolio optimization under non-Gaussian returns: mean-variance is suboptimal
- ▣ Regulatory context: Basel II $\rightarrow$ Basel III/IV framework
 - ▶ Basel II: 1% VaR with Normal assumption
 - ▶ Basel III/IV: shift from 1% VaR to 2.5% ES

Remark 1

The entire Basel regulatory capital framework rests on distributional assumptions. Underestimating tail risk $\Rightarrow$ insufficient capital $\Rightarrow$ systemic fragility.

S&P 500: What the Real World Looks Like



- Crashes ($|r_t| > 3\sigma$) occur far more frequently than the Normal distribution predicts
- Extreme returns **cluster**; we need models with heavy tails

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Normal Distribution vs. Reality — Visualization and Example

(ext. Ch. 1)

Black Monday — October 19, 1987

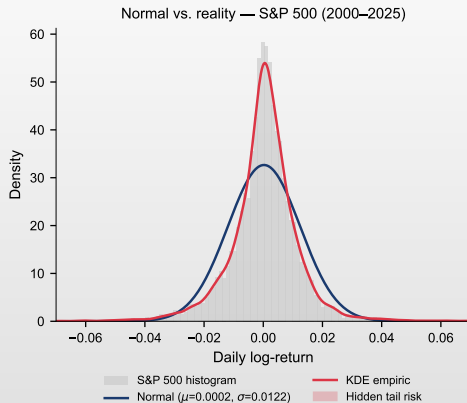
S&P 500 fell by 20.5% in a single day.

- Under the Normal assumption ($\sigma \approx 1\%$): a $\sim 20\sigma$ event with $P \approx 10^{-89}$
- Under Student- t ($\nu=4$): $P \approx 10^{-5} - 10^{84} \times$ more plausible

Return Period

- $T = \frac{1}{252 \times 10^{-89}} \approx 4 \times 10^{86}$ years; age of the universe: $\approx 1.4 \times 10^{10}$ years
- $\Rightarrow \sim 10^{76}$ **ages of the universe!**
- The distribution choice is not a technical detail — it is a **business decision**

Normal Distribution vs. Reality — Chart



SFM_ch2_normal_distribution

Model Uncertainty and Robust Risk Measures

- ▣ **Model uncertainty:** Which distribution is “correct”? Perhaps none.
- ▣ **Ambiguity aversion** (Ellsberg paradox in portfolio choice)
- ▣ Robust risk measures under distributional uncertainty:

$$\rho_{\text{robust}}(X) = \sup_{Q \in \mathcal{Q}} \rho_Q(X)$$

where $\mathcal{Q}$ is an ambiguity set of plausible distributions

- ▣ Minimax approaches: worst-case distributional analysis
- ▣ **Model risk quantification:** range of VaR/ES across plausible models provides a “model risk premium”

Definition of Log-Returns

Definition 1 (Log-Return)

The **log-return** (continuously compounded return) at time t is:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right) = \ln P_t - \ln P_{t-1}$$

where P_t is the asset price at time t .

- ▣ **Additivity:** Multi-period returns are sums: $r_{t:t+k} = \sum_{i=1}^k r_{t+i}$
- ▣ **Approximation:** For small returns, $r_t \approx (P_t - P_{t-1})/P_{t-1}$
- ▣ **Statistical motivation:** Additive structure enables standard probability theory (CLT, stable limits)

Properties of Log>Returns

Why log-returns?

- Time-additive: $r_{[0,T]} = \sum_{t=1}^T r_t$
- Symmetric: +5% and -5% are comparable
- Unbounded below: can model large losses
- Enable CLT-based inference

Simple vs. log-returns

- Simple return: $R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$
- Relationship: $r_t = \ln(1 + R_t)$
- Simple returns are *not* additive across time
- But simple returns *are* additive across portfolio assets

Convention

- Throughout this chapter, r_t denotes the **log-return** unless stated otherwise

Continuous-Time Limit

- Geometric Brownian Motion: $dS/S = \mu dt + \sigma dW$
- By Itô's lemma: $d \ln S = (\mu - \sigma^2/2) dt + \sigma dW$
- Jensen's inequality: $\mathbb{E}[\ln(P_T/P_0)] \neq \ln(\mathbb{E}[P_T/P_0])$
 - ▶ The arithmetic mean return exceeds the geometric mean return
 - ▶ Difference $\approx \sigma^2/2$ ("volatility drag")
- Log-return vs. simple return: implications for portfolio aggregation
 - ▶ Log-returns: additive across *time*
 - ▶ Simple returns: additive across *assets*
 - ▶ Neither is additive in both dimensions simultaneously

Volatility Drag — Demonstration

Thought experiment

- ▣ Your portfolio gains 50% today and loses 50% tomorrow
- ▣ Arithmetic mean return: $(+50\% - 50\%)/2 = 0\%$. But the real money?
- ▣ $\$100 \rightarrow \$150 \rightarrow \$75$. You lost **25%**!

Why log-returns tell the truth

- ▣ $r_1 = \ln(1.5) = +40.5\%$, $r_2 = \ln(0.5) = -69.3\%$
- ▣ Sum: $r_1 + r_2 = -28.8\% = \ln(0.75)$ ✓ (additivity works!)
- ▣ Simple return is **not** additive: $(+50\%) + (-50\%) \neq \ln(0.75)$
- ▣ **Rule:** $\mathbb{E}[\text{geometric return}] \approx \mathbb{E}[\text{arithmetic return}] - \sigma^2/2$; high volatility $\Rightarrow$ more severe real loss

Semimartingale Representation

- Price processes modeled as **semimartingales**: $S_t = S_0 + A_t + M_t$
- Itô's lemma for semimartingales with jumps:

$$d \ln S_t = \frac{dS_t}{S_{t-}} - \frac{1}{2} \frac{d[S, S]_t^c}{S_{t-}^2} + \sum_{s \leq t} \left(\ln \frac{S_s}{S_{s-}} - \frac{\Delta S_s}{S_{s-}} \right)$$

- Quadratic variation and realized volatility:

$$[X, X]_t = \lim_{\|\Pi\| \rightarrow 0} \sum_i (X_{t_{i+1}} - X_{t_i})^2$$

- Non-continuous price processes: jumps and Lévy–Itô decomposition
- High-frequency returns reveal microstructure noise, bid-ask bounce, Epps effect

“The Prince of Mathematicians”



Carl Friedrich Gauss
(1777–1855)

From stars to Wall Street

- ▣ Derived the normal curve in 1809 for **astronomical errors**
- ▣ The idea: errors are the sum of many small, independent perturbations
- ▣ Appears on the **10 DM banknote** — the only distribution on a banknote!

The warning

- ▣ Gauss derived it for measurement errors, **not** for financial markets
- ▣ Wall Street adopted it anyway — the consequences will be seen in this chapter

The Only Distribution on a Banknote



- 10 Deutsche Mark (series IV, 1993–2001): Gauss + normal curve with formula $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$
- On the reverse: Gauss's sextant and the geodetic triangulation network of Germany

The Normal Distribution — Definition and Properties

Definition 2 (Normal Distribution)

$X \sim N(\mu, \sigma^2)$ if:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}.$$

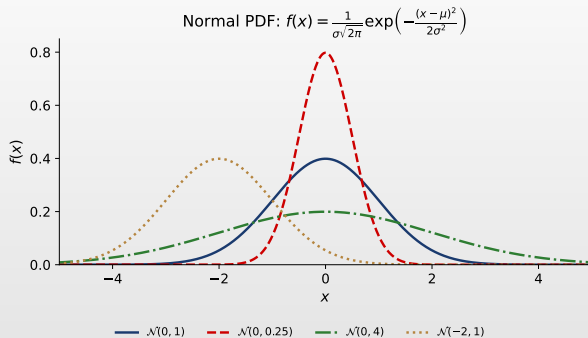
Moments:

- $\mathbb{E}[X] = \mu, \quad \text{Var}(X) = \sigma^2$
- Skewness = 0
- Kurtosis = 3 (excess kurtosis = 0)

Key properties:

- Symmetric about μ
- 68–95–99.7 rule
- Tails decay as $e^{-x^2/2}$

The Normal Distribution — PDF Chart

 SFM_ch2_normal_distribution

- **Gauss's bell:** symmetric, unimodal shape, maximum at $x = \mu$
- Parameters μ and σ^2 control the **location** and **spread**
- Tails decay **exponentially**: $f(x) \sim e^{-x^2/2}$ — underestimates extreme events

Quantiles and the Central Limit Theorem

- Quantiles: The α -quantile of $N(\mu, \sigma^2)$ is:

$$q_\alpha = \mu + \sigma \Phi^{-1}(\alpha)$$

This will be used in Section 2.10 to define Value-at-Risk

- Key quantiles: $\Phi^{-1}(0.01) = -2.326$, $\Phi^{-1}(0.05) = -1.645$

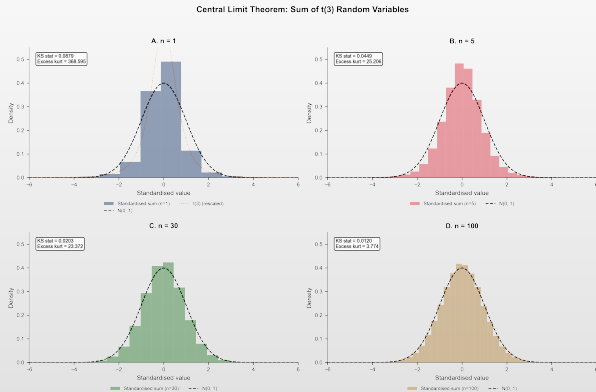
Theorem 1 (Central Limit Theorem)

If $X_1, X_2, \dots$ are i.i.d. with $\mathbb{E}[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2 < \infty$, then:

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1).$$

- Stability: Sum of independent normals is normal: $X + Y \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$

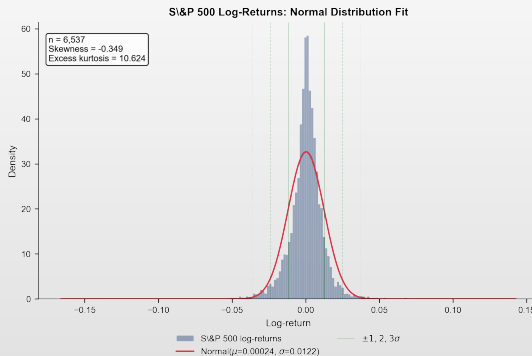
CLT Illustration



- As n increases, $\bar{X}_n$ approaches normality
- Limitation:** CLT requires finite variance — what if $\sigma^2 = \infty$?

Normal Fit to S&P 500 Returns

SFM_ch2_normal_test



- Normal PDF fits the center well but **misses the tails**
- Histogram has more mass in the tails than Normal predicts

Maximum Entropy and Normality Tests

- Maximum entropy: Among all distributions with given (μ, σ^2) , the Normal maximizes Shannon entropy

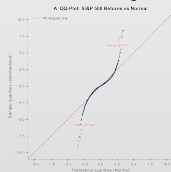
$$H = - \int f \ln f \, dx$$

- Normality tests:

Jarque–Bera test:

$$JB = \frac{n}{6} \left(S^2 + \frac{(K - 3)^2}{4} \right) \sim \chi^2(2)$$

Tests skewness S and excess kurtosis $K - 3$ jointly.



Other tests:

- Shapiro–Wilk (regression-based)
- Anderson–Darling (CDF, tail-weighted)
- Kolmogorov–Smirnov (sup CDF distance)

B. Normality Test Results

Test	Statistic	p-value (Critical)	Reject H ₀ (Yes)
Jarque-Bera	38879.71	p = 0.00e+00	Yes
Shapiro-Wilk	0.90985	p = 1.66e-47	Yes
Anderson-Darling	126.286	adj(p) = 0.7020	Yes

Sample size: n = 0.031
 Skewness: 0.34901, Kurtosis: 10.8342
 H₀: Data follows a Normal distribution

The Jarque–Bera Test

- ▣ **Maximum entropy:** Among all distributions with given (μ, σ^2) , the Normal maximizes Shannon entropy
 $H = -\int f \ln f \, dx$

Hypotheses

- H_0 : $S = 0$ and $K = 3$ (the data come from a Normal distribution)
 H_1 : $S \neq 0$ or $K \neq 3$ (departure from normality)

Test statistic

$$JB = \frac{n}{6} \left(S^2 + \frac{(K - 3)^2}{4} \right) \xrightarrow{d} \chi^2(2)$$

- ▣ $S = \frac{1}{n} \sum \left(\frac{x_i - \bar{x}}{s} \right)^3$ (skewness), $K = \frac{1}{n} \sum \left(\frac{x_i - \bar{x}}{s} \right)^4$ (kurtosis)
- ▣ Reject H_0 if $JB > \chi^2_{2, 1-\alpha}$. For $\alpha = 5\%$: $\chi^2_{2, 0.95} = 5.99$

The Shapiro–Wilk Test

Shapiro–Wilk Test (1965)

H_0 : the data come from a Normal distribution vs. H_1 : non-normality

$$W = \frac{(\sum_{i=1}^n a_i x_{(i)})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}, \quad W \in (0, 1]$$

- ▣ $x_{(i)}$ = order statistics of the sample
- ▣ a_i = tabulated coefficients, depending on n
- ▣ Reject H_0 if W is too small (close to 0)
- ▣ The most powerful test for small samples ($n < 50$)

The Anderson–Darling and Kolmogorov–Smirnov Tests

Anderson–Darling Test (1954)

$$H_0: F = F_0 \quad \text{vs.} \quad H_1: F \neq F_0$$

$$A^2 = -n - \frac{1}{n} \sum_{i=1}^n (2i-1) [\ln F_0(x_{(i)}) + \ln(1 - F_0(x_{(n+1-i)}))]$$

- Tail-weighted: more sensitive to deviations in the extremes of the distribution than KS

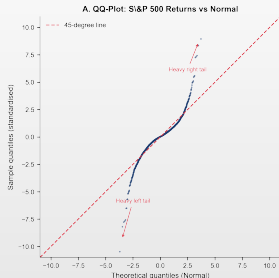
Kolmogorov–Smirnov Test (1933)

$$\square \quad H_0: F = F_0 \quad \text{vs.} \quad H_1: F \neq F_0$$

$$D_n = \sup_x |F_n(x) - F_0(x)|$$

- Non-parametric; less sensitive to tails than AD

Normality Tests — Visualization



B. Normality Test Results

Test	Statistic	p-value / Critical	Reject H ₀ (5%)
Jarque-Bera	30576.71	p = 0.00e+00	Yes
Shapiro-Wilk	0.909685	p = 1.64e-47	Yes
Anderson-Darling	126.2996	cv(5%) = 0.7520	Yes

Sample size: n = 6,537
Skewness: -0.3489 | Excess kurtosis: 10.6242
H₀: Data follows a Normal distribution

SFM_ch2_normal_distribution

- The test results on S&P 500 returns **massively** reject the normality hypothesis
- JB test: $p\text{-value} \approx 0$ (skewness and kurtosis differ from Normal values)

Normality Tests — Empirical Results

- Moment generating function: $M_X(t) = \exp(\mu t + \sigma^2 t^2 / 2)$
- Multivariate normal: $\mathbf{X} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with Mahalanobis distance

Test	Statistic	p -value	Decision
Jarque–Bera	> 1000	< 0.001	Reject H_0
Shapiro–Wilk	< 0.99	< 0.001	Reject H_0
Anderson–Darling	> 10	< 0.001	Reject H_0

Conclusion

- Normality is overwhelmingly rejected for daily financial returns
- The Normal distribution underestimates tail risk

Berry–Esseen Bounds and Convergence Rates

Theorem 2 (Berry–Esseen)

If $X_1, X_2, \dots$ are i.i.d. with $\mathbb{E}[X_i] = 0$, $\mathbb{E}[X_i^2] = \sigma^2$, $\mathbb{E}[|X_i|^3] = \rho < \infty$, then:

$$\sup_x |F_n(x) - \Phi(x)| \leq \frac{C \rho}{\sigma^3 \sqrt{n}}$$

where $C \leq 0.4748$ (Shevtsova, 2011).

- ▣ Rate of convergence: $O(n^{-1/2})$ — slow for heavy-tailed data
- ▣ Edgeworth expansions refine the CLT approximation for finite n
- ▣ **Normal as stable:** $N(\mu, \sigma^2) = S(2, 0, \sigma/\sqrt{2}, \mu)$ — the $\alpha = 2$ member of the stable family
- ▣ When $\mathbb{E}[X_i^2] = \infty$: CLT fails, but the **Generalized CLT** gives stable limits

The Pre-Asymptotic Regime — Why the CLT Does Not Save Us

The Berry–Esseen $\leftrightarrow$ aggregational Gaussianity connection

- ▣ The CLT **promises** convergence toward normality as we aggregate
- ▣ But Berry–Esseen shows the speed depends on ρ/σ^3 (third absolute moment)
- ▣ Financial returns have **very large** ρ/σ^3 (heavy tails) $\Rightarrow$ convergence is **painfully slow**

Practical implication

- ▣ If returns had thin tails, 22 days (one month) would suffice for perfect normality
- ▣ Due to heavy tails, even **annual** aggregates are not perfectly Gaussian
- ▣ **Conclusion** (Taleb): in finance, we permanently live in the *pre-asymptotic regime* — the CLT is a promise our data cannot honor

The Man Who Broke the Bell Curve



Benoît Mandelbrot
(1924–2010)

From cotton to fractals

- **1963**: cotton prices had variations *far* too large for the Normal
- Proposed **Lévy stable** distributions ($\alpha < 2$) — heavy tails, infinite variance
- Inventor of **fractal geometry**; self-similarity at every scale
- “*Markets are riskier than people think*”

The legacy

- Ignored for 40 years by Wall Street, which preferred the Gaussian model
- The 2008 crisis proved him right — thin-tailed models failed spectacularly

Fat Tails and Excess Kurtosis

- ▣ **Kurtosis:** $\kappa = \mathbb{E}[(X - \mu)^4]/\sigma^4$; excess kurtosis: $\kappa_e = \kappa - 3$
- ▣ Normal distribution: $\kappa_e = 0$
- ▣ Typical daily equity kurtosis: $\kappa \in [5, 50]$, so $\kappa_e \gg 0$
- ▣ **Interpretation:** Extreme returns occur far more often than the Normal predicts

Asset	Skewness	Excess Kurtosis
S&P 500 (daily)	-0.3	≈ 10
FTSE 100 (daily)	-0.2	≈ 8
USD/EUR (daily)	-0.1	≈ 5
Bitcoin (daily)	-0.5	≈ 15

Negative Skewness and Volatility Clustering

Negative skewness

- Left tail heavier than right tail
- Especially pronounced for equity indices
- Crashes are sharper than rallies
- Skewness $S = \mathbb{E}[(X - \mu)^3]/\sigma^3 < 0$

Volatility clustering

- “Large changes followed by large changes” (Mandelbrot, 1963)
- $\text{Corr}(|r_t|, |r_{t+k}|) > 0$ for large k
- Returns are uncorrelated but **not independent**
- $\text{Corr}(r_t, r_{t+k}) \approx 0$ but $\text{Corr}(r_t^2, r_{t+k}^2) > 0$

Implication

- A single i.i.d. distribution cannot capture volatility clustering
- We need time-varying models (GARCH, Section 2.11)

Aggregational Gaussianity

- **Key observation:** As the holding period increases, the distribution of returns becomes *more Gaussian*
- Daily returns: high kurtosis, heavy tails
- Weekly returns: moderate kurtosis
- Monthly returns: closer to Normal
- **Mechanism:** CLT-like effect — monthly return = $\sum$ of ~ 22 daily returns

Frequency	Excess Kurtosis	JB p -value
Daily	≈ 10	< 0.001
Weekly	≈ 4	< 0.001
Monthly	≈ 1.5	≈ 0.05
Quarterly	≈ 0.5	≈ 0.3

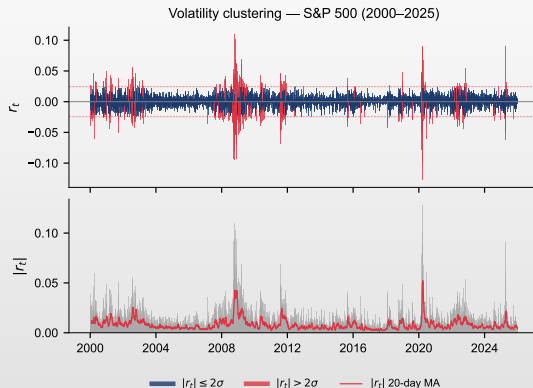
Summary of Stylized Facts

1. **Fat tails:** $\kappa_e \gg 0$ — extreme events more frequent than Normal
2. **Negative skewness:** Left tail heavier (equities)
3. **Volatility clustering:** Periods of high/low volatility persist
4. **Aggregational Gaussianity:** Kurtosis decreases with horizon
5. **No return autocorrelation:** $\text{Corr}(r_t, r_{t+k}) \approx 0$ for $k \geq 1$
6. **Squared return autocorrelation:** $\text{Corr}(r_t^2, r_{t+k}^2) > 0$

The Challenge

- Any adequate model for financial returns must reproduce these stylized facts
- The Normal distribution fails on points 1, 2, and 3

Visualizing Volatility Clustering



SFM_ch2_stylized_facts

- “Large changes followed by large changes” (Mandelbrot, 1963) — the 2001, 2008, 2020 crises
- $\text{Corr}(r_t, r_{t+k}) \approx 0$ but $\text{Corr}(|r_t|, |r_{t+k}|) > 0$ — uncorrelated, but **not independent**

Stylized Facts Equations — Kurtosis and Autocorrelation

Kurtosis

$$\kappa = \frac{\mathbb{E}[(X - \mu)^4]}{\sigma^4}$$

- ▣ **Excess kurtosis:** $\kappa_e = \kappa - 3$
- ▣ $\kappa_e = 0$: Normal; $\kappa_e > 0$: **leptokurtic**

Autocorrelation of r_t^2

$$\rho_k = \text{Corr}(r_t^2, r_{t+k}^2) > 0, \quad k \text{ large}$$

Numerical example

	κ	κ_e
Normal	3	0
S&P 500	≈ 10	≈ 7
FTSE 100	≈ 8	≈ 5
Bitcoin	≈ 15	≈ 12

- ▣ $\kappa_e = 7$ for S&P 500 $\Rightarrow$ tails **substantially heavier** than Normal

Preview: Risk Measures — VaR and ES

Definition 3 (Value-at-Risk — working definition)

Value-at-Risk at level α : the maximum loss that is **not** exceeded with probability $(1 - \alpha)$:

$$\text{VaR}_\alpha = -F^{-1}(\alpha), \quad \text{where } F^{-1} \text{ is the return quantile.}$$

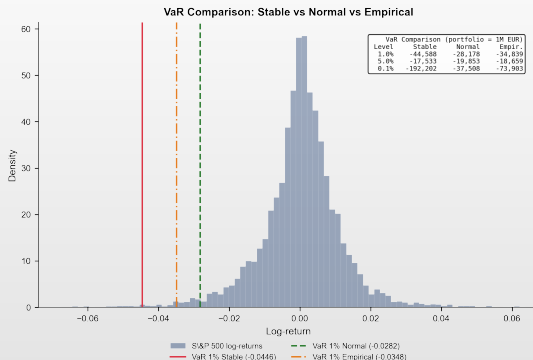
Definition 4 (Expected Shortfall — working definition)

Expected Shortfall (ES) at level α : the **average** loss in the worst $\alpha\%$ scenarios:

$$\text{ES}_\alpha = -\frac{1}{\alpha} \int_0^\alpha F^{-1}(p) dp.$$

- ▣ **Example:** Normal 1% VaR, $\sigma = 1.5\%$, \$1M: $\text{VaR} = 2.326 \times 0.015 \times 10^6 = \$34,890$
- ▣ Formal properties, Artzner coherence, backtesting — **Section 10**

VaR and ES — Graphical Illustration



- ▣ VaR_α : quantile — “we will not lose more than X with prob. $1 - \alpha$ ”
- ▣ ES_α : tail mean — “if we exceed VaR, how much do we lose *on average*?”
- ▣ $\text{ES} \geq \text{VaR}$ always; the gap grows with tail heaviness

Leverage Effect and Long Memory

Leverage effect

- ▣ Negative returns increase future volatility *more* than positive returns of equal magnitude
- ▣ Asymmetric response: $\text{Corr}(r_t, \sigma_{t+1}^2) < 0$
- ▣ Named after Black (1976): falling stock price $\Rightarrow$ higher leverage $\Rightarrow$ higher risk
- ▣ Captured by GJR-GARCH, EGARCH

Long memory in volatility

- ▣ $\text{Corr}(|r_t|, |r_{t+k}|)$ decays *slowly* (hyperbolically, not exponentially)
- ▣ Hurst exponent $H > 0.5$ for $|r_t|$
- ▣ Standard GARCH: exponential decay
- ▣ FIGARCH, HAR models capture long memory

Gain/Loss Asymmetry

- **Gain/loss asymmetry:** Stock prices decline faster than they rise
- Large drawdowns happen in days; recovery takes months or years
- This asymmetry is distinct from skewness — it concerns the *dynamics*, not the marginal distribution
- **Absence of return autocorrelation:** $\text{Corr}(r_t, r_{t+k}) \approx 0$ for $k \geq 1$
 - ▶ Markets are approximately efficient at the mean level
 - ▶ But dependence exists in higher moments ($r_t^2, |r_t|$)

Cont (2001) Taxonomy and Multifractal Models

- Cont (2001): systematic taxonomy of 11 stylized facts for asset returns
- **Multifractal models** (Mandelbrot, Fisher & Calvet, 1997):
 - ▶ $\mathbb{E}[|r_t|^q] \propto \Delta t^{\zeta(q)}$ with *nonlinear* $\zeta(q)$ (multiscaling)
 - ▶ Unifractal: $\zeta(q) = qH$ (linear); multifractal: $\zeta(q)$ concave
- Cross-correlation structure and multivariate stylized facts
- Intraday stylized facts: microstructure noise, bid-ask bounce, Epps effect

Fractal Structure and Power Laws

- ▣ **Power-law tails:** $P(|r| > x) \sim x^{-\alpha}$ with $\alpha \in [2, 5]$ for most financial assets
- ▣ Tail exponent α is remarkably stable across assets and time periods
- ▣ **Self-similarity:** Return distributions at different time scales are related by scaling
- ▣ Connection to stable distributions: if $\alpha < 2$, the GCLT implies stable limits
- ▣ The “inverse cubic law” (Gopikrishnan et al., 1999): $\alpha \approx 3$ for US equities
- ▣ Debate: true power law vs. slowly varying tails

Who Is “Student”?



William Sealy Gosset
(1876–1937)

The story behind the pseudonym

- Chemist at the **Guinness** brewery, Dublin
- The problem: how do you test beer quality from **very small** samples? ($n = 3-4$)
- Guinness **forbade** publication — trade secret!
- The solution: publish under the pseudonym “**Student**” (1908)

Why does it matter for finance?

- The Normal with σ *estimated* is no longer Normal — it is Student- t !
- The irony: invented for beer, now it models stock market crashes

Student- t Distribution — DefinitionDefinition 5 (Student- t Distribution)

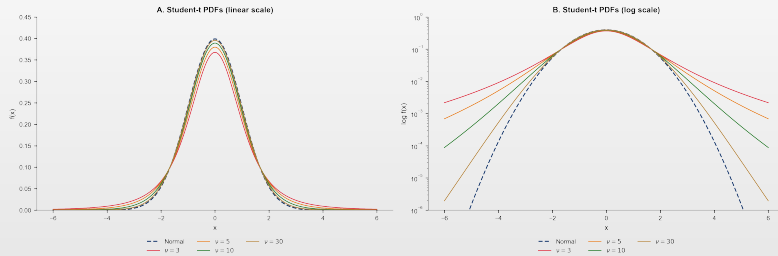
$X \sim t_\nu$ if:

$$f(x; \nu) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi} \Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{x^2}{\nu}\right)^{-(\nu+1)/2}, \quad x \in \mathbb{R}.$$

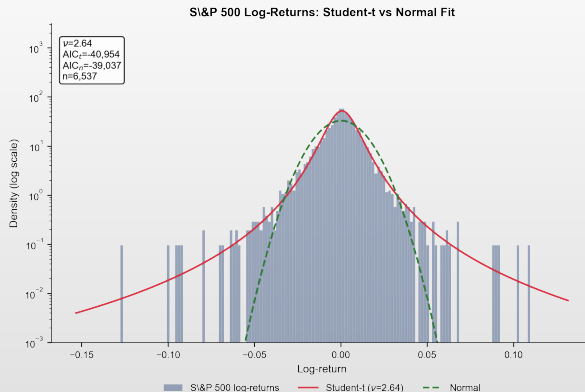
- ▣ Single parameter $\nu > 0$ (**degrees of freedom**) controls tail heaviness
- ▣ $\mathbb{E}[X] = 0$ ($\nu > 1$); $\text{Var}(X) = \frac{\nu}{\nu-2}$ ($\nu > 2$)
- ▣ Excess kurtosis: $\kappa_e = \frac{6}{\nu-4}$ ($\nu > 4$)
- ▣ $\nu \rightarrow \infty$: converges to $N(0, 1)$; $\nu = 1$: Cauchy distribution
- ▣ Tails: $f(x) \sim |x|^{-(\nu+1)}$ (**power law**)

Student- t PDFs — Effect of ν

SFM_ch2_student_t_fit



- Smaller $\nu \Rightarrow$ heavier tails, higher peak
- Financial returns: typically $\hat{\nu} \in [3, 8]$ for daily equity data

Student- t Fit to S&P 500

- Student- t captures the heavy tails much better than the Normal
- Estimated $\hat{\nu} \approx 4-5$ for daily S&P 500; center and tails well-matched

Why Student- t for Finance?

Property	Normal	Student- t
Tail decay	Exponential ($e^{-x^2/2}$)	Power law ($ x ^{-\nu-1}$)
Excess kurtosis	0	$6/(\nu - 4)$
Parameters	2 (μ, σ)	3 (μ, σ, ν)
Extreme events	Very rare	More frequent
Limit	—	Normal ($\nu \rightarrow \infty$)

Key Advantage

- Student- t **nests** the Normal as a special case ($\nu \rightarrow \infty$) while allowing heavier tails
- It is the simplest heavy-tailed alternative

Location-Scale Form and MLE

▣ **Location-scale form:** $X = \mu + \sigma T$, where $T \sim t_\nu$

▣ MLE estimation of (μ, σ, ν) : maximize

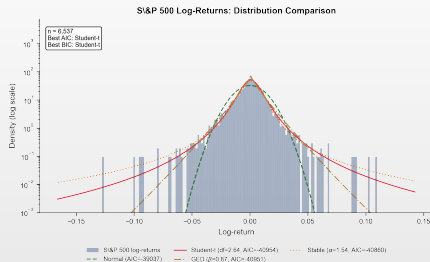
$$\ell(\mu, \sigma, \nu) = \sum_{t=1}^n \ln f\left(\frac{r_t - \mu}{\sigma}; \nu\right) - n \ln \sigma$$

▣ Information criteria for model selection:

- ▶ AIC = $-2\ell + 2k$ (penalizes complexity lightly)
- ▶ BIC = $-2\ell + k \ln n$ (penalizes complexity more)

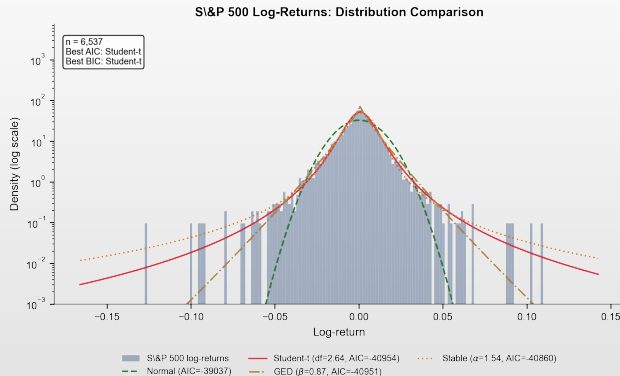
▣ Student- t as innovation distribution in GARCH models: $\epsilon_t = \sigma_t z_t$, $z_t \sim t_\nu$

▣ QQ-plot diagnostics: departures from the 45° line indicate misspecification

Model Comparison: Normal vs. Student- t 

Model	Log-lik	AIC	BIC
Normal (μ, σ)	ℓ_1	$-2\ell_1 + 4$	$-2\ell_1 + 2 \ln n$
Student- t (μ, σ, ν)	$\ell_2 > \ell_1$	$-2\ell_2 + 6$	$-2\ell_2 + 3 \ln n$

- Student- t achieves substantially better AIC/BIC despite one extra parameter

Normal vs. Student- t — Overlay on S&P 500 **SFM_ch2_distribution_comparison**

- Student- t captures the heavy tails; the Normal systematically underestimates them

Model Comparison: AIC/BIC

Model	Log-lik.	AIC	BIC
Normal (μ, σ)	ℓ_1	$-2\ell_1 + 4$	$-2\ell_1 + 2 \ln n$
Student- t (μ, σ, ν)	$\ell_2 > \ell_1$	$-2\ell_2 + 6$	$-2\ell_2 + 3 \ln n$

- Student- t achieves substantially better AIC/BIC despite one extra parameter
- The log-likelihood gain comes from the superior tail fit

Mixture Representation and Bayesian Interpretation

- **Mixture representation:** $X | V \sim N(0, V)$, $V \sim \text{Inv-Gamma}(\nu/2, \nu/2)$
 - ▶ Student- t = Normal with random variance (scale mixture of normals)
 - ▶ Interpretation: V captures time-varying volatility
- **Bayesian interpretation:** Student- t as marginal posterior under conjugate normal-inverse-gamma prior
- **Multivariate Student- t :** $f(\mathbf{x}) \propto \left(1 + \frac{(\mathbf{x}-\boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{x}-\boldsymbol{\mu})}{\nu}\right)^{-(\nu+p)/2}$
- **Tail comparison:** t_ν tail index = ν vs. stable tail index = α
- Student- t is **not** closed under convolution; stable distributions **are**

Model Uncertainty and Robust Risk Measures

- ▣ **Model uncertainty:** Which distribution is “correct”? Perhaps none — every model is an *approximation*.
- ▣ **Ambiguity aversion** (Ellsberg paradox): the decision-maker does not know the exact distribution and adopts a conservative attitude.
- ▣ Robust risk measures under distributional uncertainty:

$$\rho_{\text{robust}}(X) = \sup_{Q \in \mathcal{Q}} \rho_Q(X)$$

- ▣ **Ambiguity set** $\mathcal{Q}$ = a set of probability distributions deemed plausible. We compute risk under the *worst-case* distribution from $\mathcal{Q}$.
- ▣ Classic examples of $\mathcal{Q}$:
 - ▶ **Moment-based:** all distributions with given (μ, σ^2) : $\mathcal{Q} = \{Q : \mathbb{E}_Q[X] = \mu, \text{Var}_Q(X) = \sigma^2\}$
 - ▶ **Wasserstein:** neighborhood of radius ε around the empirical distribution: $\mathcal{Q} = \{Q : W_p(Q, \hat{P}_n) \leq \varepsilon\}$
 - ▶ **KL-divergence:** $\mathcal{Q} = \{Q : D_{\text{KL}}(Q \| P_0) \leq \eta\}$
- ▣ **Model risk quantification:** the VaR/ES range across plausible models provides a “model risk premium”

Distances Between Distributions — Intuition

- **Problem:** How do we measure how “close” two distributions P and Q are?
- **Wasserstein distance** $W_p(P, Q)$:
 - ▶ Also known as *Earth Mover's Distance*
 - ▶ Intuition: the minimum cost of “transporting” probability mass from P to Q
 - ▶ If P and Q are distributions on $\mathbb{R}$ with CDFs F_P, F_Q :

$$W_1(P, Q) = \int_{-\infty}^{+\infty} |F_P(x) - F_Q(x)| dx$$

- ▶ Advantage: accounts for the *geometry* of the space (not just density overlap)
- **Kullback–Leibler divergence** $D_{\text{KL}}(Q \| P_0)$:
 - ▶ Measures the “information loss” when approximating Q by P_0
 - ▶ $D_{\text{KL}}(Q \| P_0) = \int q(x) \ln \frac{q(x)}{p_0(x)} dx \geq 0$, with equality iff $Q = P_0$
 - ▶ **Not** a proper distance (not symmetric: $D_{\text{KL}}(Q \| P) \neq D_{\text{KL}}(P \| Q)$)
- **Application:** The choice of radius ε (Wasserstein) or η (KL) controls the *degree of conservatism* of the robust risk measure — larger $\varepsilon \Rightarrow$ wider $\mathcal{Q} \Rightarrow$ higher robust VaR

Wasserstein in Practice — Bank Stress-Testing

Scenario: The 2008 Crisis / 2020 Pandemic

- ▣ The *historical* return distribution suddenly becomes irrelevant (non-stationarity)
- ▣ VaR estimated on 2005–2007 data massively underestimates losses in 2008
- ▣ **Problem:** how do we protect ourselves when the model we estimated may be wrong?

Solution: robust optimization with Wasserstein neighborhood

- ▣ Instead of computing VaR under a *single* distribution $\hat{P}$, compute under the **worst-case** distribution in a neighborhood: $\rho_{\text{robust}} = \sup_{Q: W_1(Q, \hat{P}) \leq \varepsilon} \text{VaR}_Q$
- ▣ The radius ε = the bank's uncertainty budget (how much reality can deviate)
- ▣ Small ε : optimism ("the crisis will not repeat"); large ε : conservatism (survival)
- ▣ **Basel III+**: regulators mandate *stressed VaR* — exactly this philosophy

From Student- t to Pareto — Why Do We Need Both?

Student- t : the **complete** model

- ▣ Models the **entire dataset** (center + tails)
- ▣ Trade-off: fitting the center well may miss the extreme tail
- ▣ Global parameter: ν simultaneously controls the peak and tails

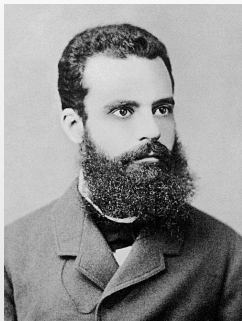
Pareto: the **extreme tail** model

- ▣ Models **only the extreme values** ($x > x_m$)
- ▣ No compromises: focuses exclusively on the right tail
- ▣ Mentally prepares for Extreme Value Theory (EVT)

The mathematical link

- ▣ Student- t with ν d.f. has **Pareto tails**: $P(X > x) \sim C \cdot x^{-\nu}$ for $x \rightarrow \infty$
- ▣ The Pareto tail index α = the degrees of freedom ν of Student- t
- ▣ Pareto is not an alternative to Student- t , but a **complementary** diagnostic tool

The 80/20 Rule — The Man Behind the Law



Vilfredo Pareto
(1848–1923)

From gardening to inequality

- ▣ Italian-Swiss engineer: **80% of Italy's land** was owned by **20% of the population** (1896)
- ▣ Legend: 20% of plants produced 80% of peas
- ▣ The same law appears **everywhere**: wealth, cities, earthquakes, sales, web traffic

Why does it matter for finance?

- ▣ A few catastrophic days generate most of the risk
- ▣ Mandelbrot (1963): return tails are Pareto — $P(X > x) \propto x^{-\alpha}$

The Pareto Distribution — Power Law

Definition (Pareto, 1896)

A random variable X follows a **Pareto distribution** with parameters $x_m > 0$ (scale) and $\alpha > 0$ (shape / tail index) if:

$$f(x) = \frac{\alpha x_m^\alpha}{x^{\alpha+1}}, \quad x \geq x_m$$

▣ **CDF:** $F(x) = 1 - \left(\frac{x_m}{x}\right)^\alpha$

▣ **Survival function:**

$$\bar{F}(x) = P(X > x) = \left(\frac{x_m}{x}\right)^\alpha$$

▣ **Power-law** behavior: $P(X > x) \propto x^{-\alpha}$

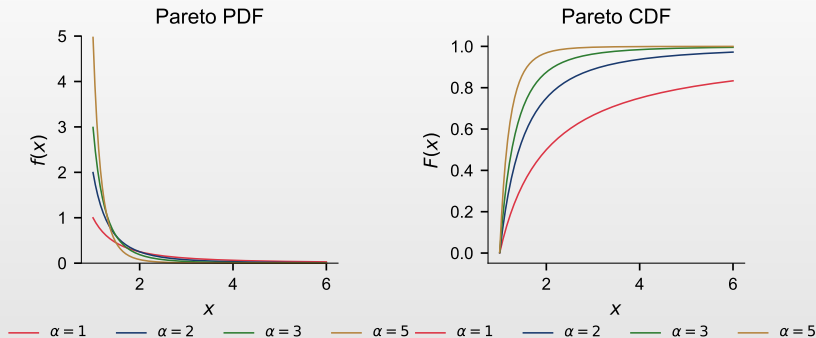
▣ **Mean:** $\mathbb{E}[X] = \frac{\alpha x_m}{\alpha - 1}$ (exists only if $\alpha > 1$)

▣ **Variance:** $\text{Var}(X) = \frac{\alpha x_m^2}{(\alpha - 1)^2(\alpha - 2)}$ (exists only if $\alpha > 2$)

▣ $\alpha \leq 2$: **infinite variance**

▣ $\alpha \leq 1$: **infinite mean**

The Pareto Distribution — PDF and CDF

[SFM_ch2_pareto_analysis](#)

- Small $\alpha \Rightarrow$ heavy tail, density decays slowly
- Large $\alpha \Rightarrow$ mass concentrates near x_m , tail decays rapidly

The Pareto Distribution — Power-Law Signature

Q SFM_ch2_pareto_analysis

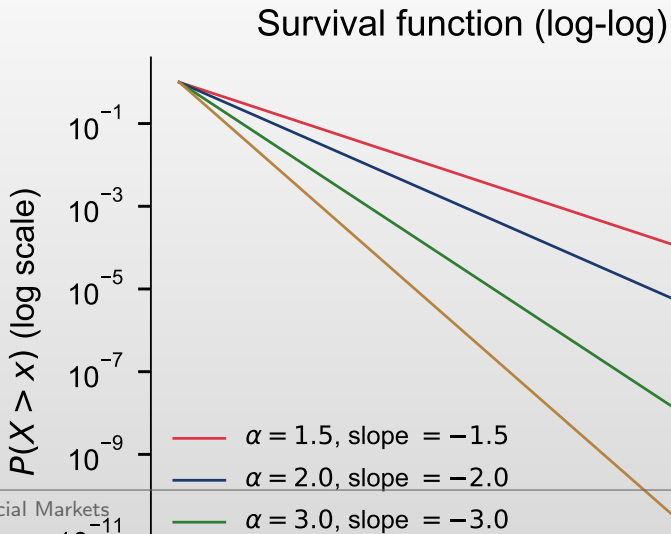
Log-log graphical test

- On a log-log plot, the Pareto survival function is a **straight line**:

$$\log P(X > x) = -\alpha \log x + \alpha \log x_m$$

- **Slope** = $-\alpha$ (tail index)
- Alignment on a line $\Rightarrow$ Pareto behavior; deviation for small x : *truncation*
- Practical check: plot $\log \hat{S}(x)$ vs. $\log x$ on empirical data — linearity confirms a Pareto tail

The Pareto Distribution — Log-log Survival



The Pareto Distribution — Why It Matters in Finance

The Pareto principle (80/20 rule)

- ▣ **Vilfredo Pareto** (1896): 80% of Italy's wealth was held by 20% of the population
- ▣ The same law appears in finance: extreme losses follow a power law
- ▣ **Mandelbrot** (1963): market returns have Pareto-type tails

Key property: heavy tails

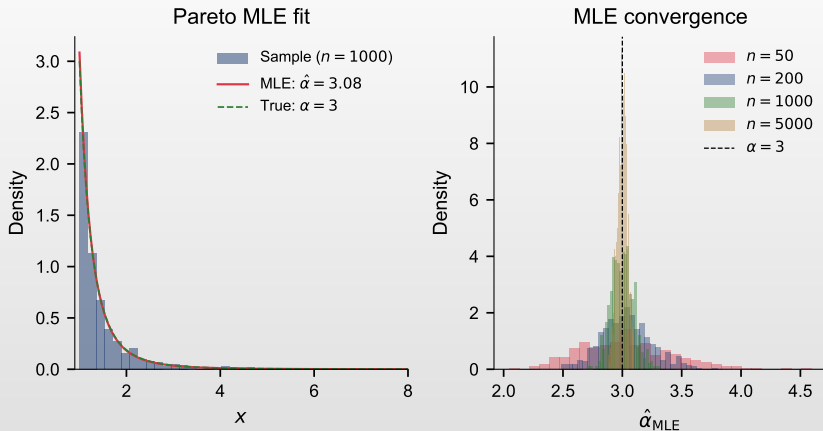
- ▣ Normal: $P(X > x) \sim e^{-x^2/2}$ (**exponential**); Pareto: $P(X > x) \sim x^{-\alpha}$ (**polynomial**)
- ▣ *Black Swan* events are **much more probable** under Pareto
- ▣ **Example:** $\alpha = 3$, $P(X > 10x_m) = 10^{-3}$; under Normal: $\approx 10^{-23}$

The Pareto Distribution — Parameter Estimation (MLE)

Maximum Likelihood Estimator (MLE)

- ▣ Let $x_1, \dots, x_n$ be an i.i.d. sample from $\text{Pareto}(\alpha, x_m)$
- ▣ **Step 1:** $\hat{x}_m = \min(x_1, \dots, x_n)$ (scale = minimum value)
- ▣ **Step 2:** $\hat{\alpha}_{\text{MLE}} = \frac{n}{\sum_{i=1}^n \ln(x_i / \hat{x}_m)}$
- ▣ **Properties:** consistent, asymptotically efficient, $\sqrt{n}$ -convergent

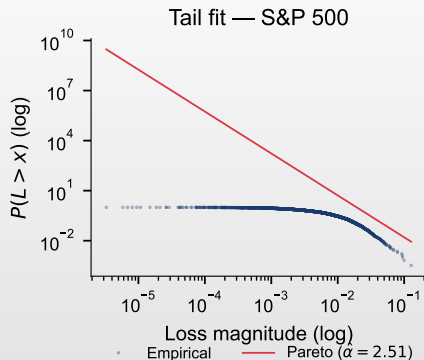
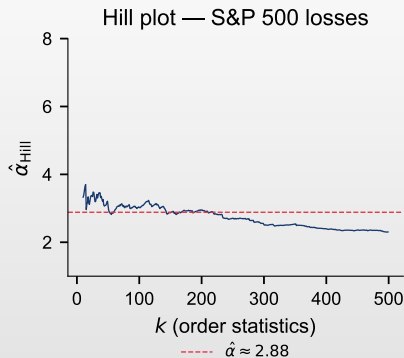
The Pareto Distribution — MLE: Demonstration



□ **Left:** histogram + MLE fit vs. true value ($\alpha = 3$, $n = 1000$)

□ **Right:** MLE convergence as n increases ($n = 50 \rightarrow 5000$)

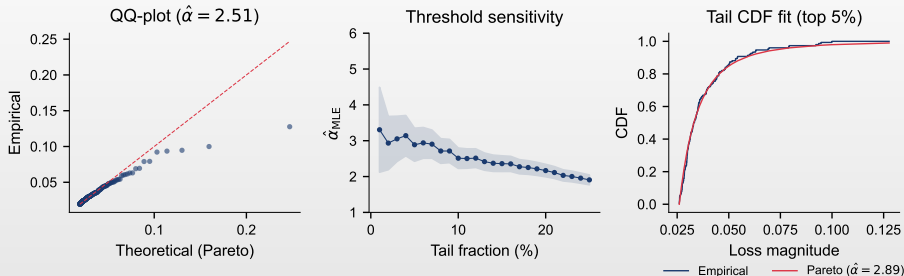
The Pareto Distribution — Hill Estimator on S&P 500



SFM_ch2_pareto_analysis

- **Hill plot:** stable region $\hat{\alpha} \approx 2.9$; **Tail fit:** Pareto $\hat{\alpha} \approx 2.5$
- $\alpha < 3 \Rightarrow$ moments of order ≥ 3 do not exist; confirmation: **heavy tails**

The Pareto Distribution — Tail Fitting (S&P 500)



 SFM_ch2_pareto_analysis

- **QQ-plot:** alignment on diagonal $\Rightarrow$ tail is Pareto
- **Threshold sensitivity:** stable region $\approx 3\text{--}8\%$; $\hat{\alpha} \approx 2.5\text{--}3.0$

The Pareto Distribution — Tail Fitting Interpretation

Tail fitting procedure

- ▣ **Step 1:** Select threshold u (or tail fraction, e.g. top 5%)
- ▣ **Step 2:** Estimate $\hat{\alpha}$ on data $\{x_i : x_i > u\}$ via MLE or Hill
- ▣ **Step 3:** Check sensitivity: vary u and observe $\hat{\alpha}$ stability
- ▣ **Step 4:** Validate with QQ-plot and CDF fit

Empirical results (S&P 500, 2000–2025)

- ▣ $\hat{\alpha} \approx 2.5\text{--}3.0 \Rightarrow$ finite moments only up to order 2
- ▣ Variance exists, but kurtosis **does not** — the Normal distribution is inadequate
- ▣ The Pareto CDF fits well on the right tail of losses

The Pareto Distribution — Applications

 SFM_ch2_pareto_analysis

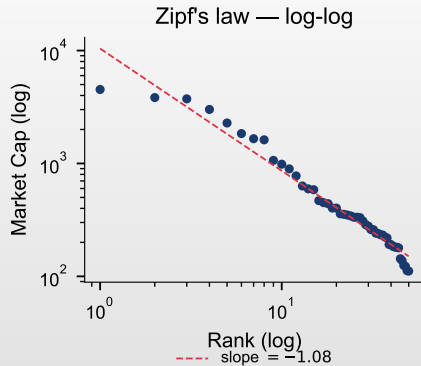
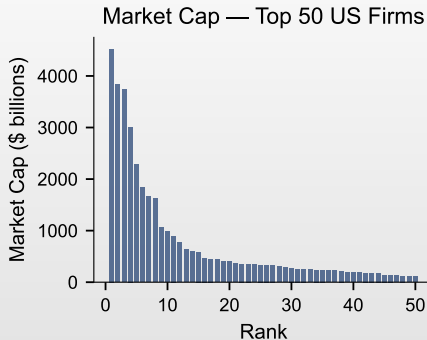
Zipf's law (market capitalization)

- ▣ Firm capitalization follows a **power law**: $P(\text{cap} > x) \propto x^{-\alpha}$
- ▣ Top 10 firms $\approx$ 30% of the market; Zipf slope ≈ -1 (universal empirical law)

Other applications of the Pareto distribution

- ▣ **Insurance losses**: extreme insured amounts $\Rightarrow$ Pareto
- ▣ **Operational risk** (Basel II): operational losses have Pareto tails
- ▣ **Wealth distribution**: Top 1% holds 40–50% of wealth (Piketty, 2014)
- ▣ **Trading volume**: a few large trades dominate total volume
- ▣ **Cyber risk**: severity of cyber attacks is Pareto-distributed

Zipf's Law — Market Capitalization



The Pareto Distribution — Theoretical Connections

Connection with EVT and the Hill estimator

- ▣ **Pickands–Balkema–de Haan theorem:** exceedances above a sufficiently high threshold approximately follow a **Generalized Pareto Distribution (GPD)**
- ▣ The classical Pareto distribution = special case of GPD with $\xi > 0$
- ▣ **Hill estimator:** estimates α from order statistics $X_{(1)} \leq \dots \leq X_{(n)}$:

$$\hat{\alpha}_{\text{Hill}} = \left(\frac{1}{k} \sum_{i=1}^k \ln \frac{X_{(n-i+1)}}{X_{(n-k)}} \right)^{-1}$$

Connection with stable distributions and Student- t

- ▣ α -stable distributions with $\alpha < 2$ have Pareto tails: $P(X > x) \sim C_{\alpha} x^{-\alpha}$
- ▣ Student- t with ν degrees of freedom: Pareto tails with index $\alpha = \nu$
- ▣ The Pareto tail index unifies the theory: smaller $\alpha \Rightarrow$ heavier tails

Case Study: The Data

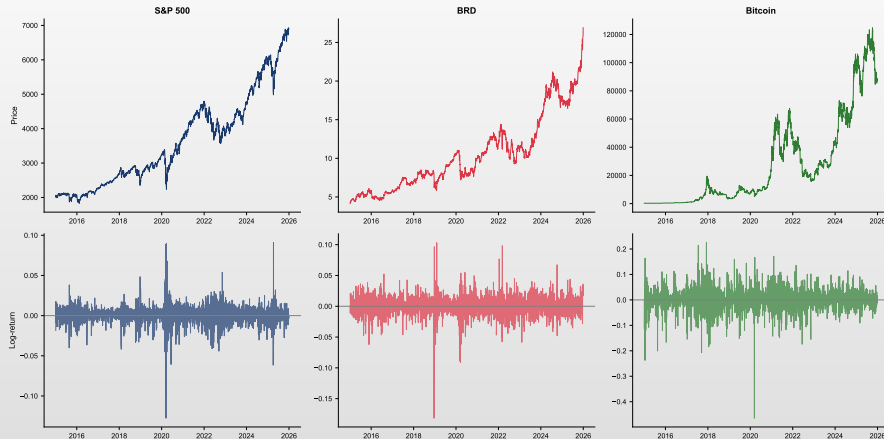
SFM_ch2_case_study_2a

Prices and log-returns

- ▣ **S&P 500** — developed market, high liquidity
- ▣ **BRD** — Romanian stock, emerging market
- ▣ **Bitcoin** — digital asset, extreme volatility
- ▣ Period: 2015–2025, daily data

Asset	N	Mean (%)	Std (%)	Skew	Kurt
S&P 500	~2 500	0.05	1.15	−0.60	13.5
BRD	~2 500	0.04	1.10	−0.80	12.0
Bitcoin	~3 600	0.12	3.80	−0.30	8.5

Case Study: Prices and Log-Returns



Case Study: Stylized Facts

 SFM_ch2_case_study_2a

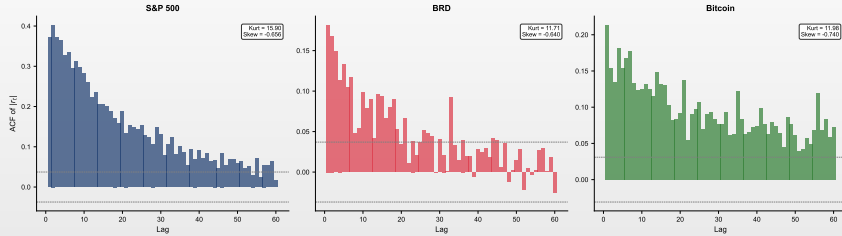
Volatility clustering

- ▣ ACF of $|r_t|$ confirms long-range dependence
- ▣ All 3 assets exhibit significant autocorrelation of $|r_t|$ up to lag 60
- ▣ **Kurtosis** $\gg 3$ across all markets $\Rightarrow$ heavy tails
- ▣ **Negative skewness** for S&P 500 and BRD (*leverage effect*)
- ▣ Bitcoin: lower skewness, but $3\times$ higher volatility

Remark 2

- ▣ Volatility clustering invalidates the i.i.d. assumption required by the Black–Scholes model

Case Study: Stylized Facts — Chart



 SFM_ch2_case_study_2a



Case Study: Normal vs Student- t

SFM_ch2_case_study_2a

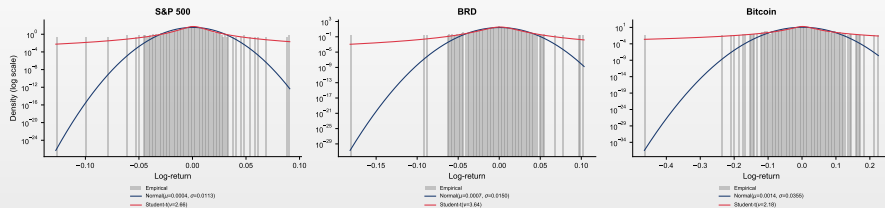
Log-scale histogram with overlay

- We compare the two models on a logarithmic scale
- Normal systematically underestimates the tails
- Student- t adapts well with estimated ν :
 - ▶ S&P 500: $\hat{\nu} \approx 4-5$; BRD: $\hat{\nu} \approx 3-5$; Bitcoin: $\hat{\nu} \approx 3-4$
- The difference grows exponentially in the tails

Implication

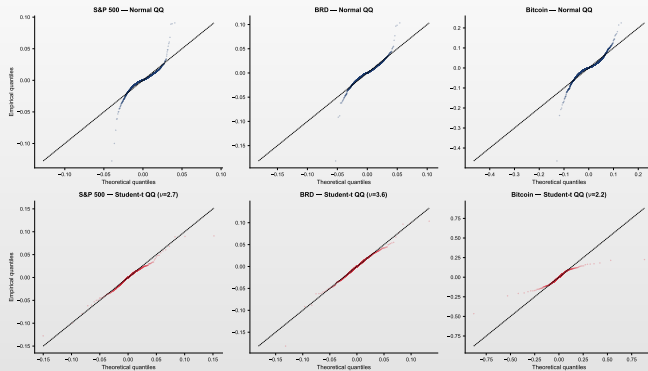
- On a logarithmic scale, the Normal underestimation becomes visible: empirical tails are orders of magnitude heavier

Case Study: Normal vs Student- t — Chart



SFM_ch2_case_study_2a

Case Study: QQ-Plots



SFM_ch2_case_study_2a

- **Normal QQ:** systematic deviations in tails **Student-t QQ:** much better alignment
- **Bitcoin:** largest deviations, confirming heavier tails

Case Study: Normality Tests

Testing H_0 : returns $\sim \mathcal{N}(\mu, \sigma^2)$

Asset	Jarque–Bera	Shapiro–Wilk	Anderson–Darling	Kolmogorov–Smirnov
S&P 500	> 10 000***	< 0.99***	> 20***	> 0.05***
BRD	> 8 000***	< 0.99***	> 15***	> 0.04***
Bitcoin	> 5 000***	< 0.98***	> 30***	> 0.06***

□ *** rejection at level $\alpha = 0.001$ for all tests and all assets

Conclusion

- Normality is **decisively rejected** across all three markets
- All four tests converge: JB (moments), SW (order), AD (tails), KS (full distribution)

Case Study: VaR and ES — Normal vs Student- t Applying risk formulas (portfolio 1M EUR, $\mu \approx 0$)

$$\square \text{ VaR}_{1\%}^N = 2.326 \sigma; \quad \text{VaR}_{1\%}^t = \sqrt{\frac{\nu-2}{\nu}} |t_{\nu}^{-1}(0.01)| \sigma; \quad \text{ES: left-tail mean below VaR}$$

Asset ($\sigma, \hat{\nu}$)	VaR Normal	VaR Student- t	ES Normal	ES Student- t
S&P 500 (1.15%, 4.5)	26 750	30 400	30 500	40 200
BRD (1.10%, 3.8)	25 590	29 600	29 300	40 200
Bitcoin (3.80%, 3.2)	88 390	102 000	100 900	148 700

Normal ES underestimates by 30–50%

$$\square P(r < -3\sigma): \text{Normal} = \Phi(-3) = 0.135\% \text{ (1 day/741); } \text{Student-}t \text{ } (\hat{\nu}=4.5) \approx 0.58\% \text{ (1 day/172)} \text{ — } 4.3\times \text{ more frequent}$$

Case Study: Model Selection — AIC/BIC

Information criteria ($AIC = -2\ell + 2k$; $BIC = -2\ell + k \ln n$)

Asset	ΔAIC (Normal – Student- t)	ΔBIC	Winner
S&P 500	> 200	> 190	Student- t
BRD	> 150	> 140	Student- t
Bitcoin	> 300	> 290	Student- t

Remark 3

- ▣ $\Delta AIC > 10 \Rightarrow$ strong evidence against the Normal model (Burnham & Anderson)
- ▣ Student- t wins decisively across all markets, despite having one extra parameter (ν)

Case Study: Conclusions

Lessons from real data analysis

1. **Kurtosis** $\gg 3$ across all markets $\Rightarrow$ the Normal model is inadequate
2. **Student- t** provides a first approximation, but has *symmetric tails*
3. **Skewness** (especially for S&P 500 and BRD) requires asymmetric models
4. **Volatility clustering** demands conditional models (GARCH)

Risk management implications

- ▣ Normal ES at 1% underestimates the mean tail loss by **30–50%** compared to Student- t
- ▣ Normal VaR underestimates by $\sim 15\%$, but 3σ events occur $4\times$ more often

AI Exercise: Distributions and Tail Risk

Prompt to test in ChatGPT / Claude / Copilot

"Download daily data for S&P 500 and Bitcoin from 2015 to 2025 using Python (yfinance). Compute log-returns. Fit a Normal and a Student- t distribution via MLE on each series. Compare models using AIC/BIC, QQ-plot, and log-scale histogram. Compute 1% VaR and Expected Shortfall under both models. Estimate the Pareto tail index using the Hill estimator."

Exercise:

1. Does the AI compute log-returns correctly? Check: $r_t = \ln(P_t/P_{t-1})$, not $(P_t - P_{t-1})/P_{t-1}$
2. Are the estimated degrees of freedom $\hat{\nu}$ plausible ($\hat{\nu} \in [3, 8]$ for daily data)?
3. Are the VaR and ES formulas correct? Check: $\text{VaR}_{1\%}^t$ uses adjusted scale $s = \sigma\sqrt{(\nu - 2)/\nu}$?
4. Ask the AI: "What happens to VaR if $\nu \leq 2$?" Does it mention infinite variance?
5. Does the Hill plot show a stable region? What value of $\hat{\alpha}$ does it obtain? Compare with $\hat{\nu}$ from Student- t

☐ **Warning:** AI often confuses *excess kurtosis* ($\kappa - 3$) with total kurtosis (κ). Always verify the definition used.

Question 1

Question

- The event of October 19, 1987 (S&P 500 -20.5%) corresponds to a $\sim 20\sigma$ event under the Normal assumption ($\sigma \approx 1\%$). What is the probability of such an event?

Answer choices

- (A) $\approx 10^{-2}$ (1 in 100)
- (B) $\approx 10^{-30}$ (practically impossible on a human timescale)
- (C) $\approx 10^{-89}$ (impossible in the lifetime of the universe)
- (D) $\approx 10^{-6}$ (once every million days)

Question 1: Answer

Answer: (C)

- ▣ **Normal:** $\sigma \approx 1\%$, so $-20.5\% = 20.5\sigma$. $P = 2\Phi(-20.5) \approx 10^{-89}$.
- ▣ **Student- t ($\nu = 5$):** adjusted scale $s = \sigma\sqrt{3/5} = 0.775\%$; $t_{\text{stat}} = 20.5/0.775 = 26.5$; $P(|T_5| > 26.5) \approx 10^{-8}$ — still extreme, but $10^{81} \times$ **more probable** than under Normal!
- ▣ Yet it happened — demonstrating the **inadequacy of the Normal distribution**.
- ▣ Under stable distributions ($\alpha \approx 1.7$), the probability rises to $\sim 10^{-3}$.

Question 2

Question

- ☐ A Student- t distribution with $\nu = 4$ degrees of freedom has:

Answer choices

- ☐ (A) Finite mean, finite variance, finite kurtosis
- ☐ (B) Finite mean, finite variance, infinite kurtosis
- ☐ (C) Finite mean, infinite variance, infinite kurtosis
- ☐ (D) Infinite mean, infinite variance, infinite kurtosis

Question 2: Answer

Answer: (B)

- ▣ For Student- t : mean exists if $\nu > 1$, variance if $\nu > 2$, kurtosis if $\nu > 4$.
- ▣ With $\nu = 4$: finite mean ($\checkmark$), finite variance ($= \nu/(\nu - 2) = 2$, $\checkmark$), but kurtosis **does not exist** (4th moment diverges: $6/(\nu - 4)$ with $\nu = 4$).
- ▣ (A) is wrong: the 4th moment does not exist at $\nu = 4$ exactly.
- ▣ This shows that even distributions with finite variance can have **very heavy** tails.

Question 3: Reasoning — Basel III and Distribution Choice

Question

- ▣ A bank computes regulatory capital using ES at 2.5% under the Normal distribution. Supervisors discover that portfolio returns have $\hat{\nu} = 4$ (Student- t). What is the consequence?

Answer choices

- ▣ **(A)** Capital is sufficient — ES does not depend on the distribution shape
- ▣ **(B)** Capital is underestimated by 30–50%, the bank must increase reserves
- ▣ **(C)** Capital is overestimated — Student- t predicts lower risks
- ▣ **(D)** Cannot decide without knowing the volatility

Question 3: Answer

Answer: (B)

- ▣ Student- t with $\nu = 4$ has much heavier tails than Normal: $P(r < -3\sigma)$ is $\sim 4\times$ larger.
- ▣ ES under Student- t is **30–50% larger** than under Normal (as demonstrated in the case study).
- ▣ (A) is wrong: ES depends critically on the shape of the tails, not just on σ .
- ▣ (D) is wrong: even with the same σ , Student- t gives larger ES due to heavier tails.
- ▣ **Lesson:** Distribution choice is not academic — it **directly determines bank capital**.

Exercise 1: Log>Returns and Moments

Problem

□ Given daily closing prices:

t	0	1	2	3	4
P_t	100	103	98	95	102

1. Compute log-returns $r_t = \ln(P_t/P_{t-1})$ for $t = 1, \dots, 4$.
2. Compute sample mean, variance, skewness, and kurtosis.
3. Apply the Jarque–Bera test: $JB = \frac{n}{6} \left(S^2 + \frac{(K-3)^2}{4} \right)$. Is normality rejected at 5%?
4. Compute the total log-return over 4 periods. Verify time-additivity.

Exercise 2: The Normal Distribution and VaR

Problem

□ A \$5M portfolio has daily returns $\sim N(\mu, \sigma^2)$ with $\hat{\mu} = 0.04\%$ and $\hat{\sigma} = 1.2\%$

1. Compute daily VaR at levels 5%, 1%, and 0.1%.
2. Compute Expected Shortfall at the 1% level.
3. A -4% event occurred. What is its probability under the Normal assumption?
4. Argue why Normal VaR may underestimate real risk.

Hint

$\text{VaR}_\alpha = -(\mu + z_\alpha \cdot \sigma)$, where $z_{0.05} = -1.645$, $z_{0.01} = -2.326$, $z_{0.001} = -3.090$.

Exercise 3: Student- t — Fitting and Comparison

Problem

▣ Using the data from Exercise 1, suppose MLE produces: Normal $\ell = 12.5$, Student- t ($\nu = 5$) $\ell = 14.8$

1. Compute AIC and BIC for both models ($k_N = 2$, $k_t = 3$, $n = 4$).
2. Which model is preferred by AIC? By BIC?
3. Compute the excess kurtosis of the Student- t distribution with $\nu = 5$.
4. Explain why Student- t captures heavy tails better than the Normal distribution.

Formulas

▣ $\text{AIC} = -2\ell + 2k$, $\text{BIC} = -2\ell + k \ln n$, $\kappa = 6/(\nu - 4)$ for $\nu > 4$

Exercise 4: Comparing VaR/ES — Normal, Student- t , Pareto

Problem

□ A \$10M portfolio has daily returns with $\hat{\sigma} = 1.5\%$, $\mu \approx 0$. The Hill estimator on the left tail gives $\hat{\alpha} = 3.5$

1. Compute 1% VaR under: (a) Normal; (b) Student- t with $\nu = 5$; (c) Pareto with $\alpha = 3.5$, $x_m = 0.01$
2. Compute 1% ES under each model. *Hint:* $ES_{\alpha}^{\text{Pareto}} = \frac{\alpha}{\alpha-1} \text{VaR}_{\alpha}$ for $\alpha > 1$
3. Fill in the table and discuss the differences:

	Normal	Student- t ($\nu=5$)	Pareto ($\alpha=3.5$)
VaR 1%	?	?	?
ES 1%	?	?	?
ES/VaR	?	?	?

Python Lab — Interactive Exercises

Lab 1: Distribution Fitting (Y3)

1. Open `SFM_ch2_normal_distribution.ipynb`
2. Download S&P 500 data (2015–2025); compute log-returns
3. Histogram + Normal overlay; Jarque–Bera test
4. Fit Student- t (MLE). Compare AIC with Normal model

 `SFM_ch2_normal_distribution`

Exercise 1: Solution

Solution

1. Log-returns: $r_1 = \ln(103/100) = 0.0296$, $r_2 = \ln(98/103) = -0.0498$, $r_3 = \ln(95/98) = -0.0311$, $r_4 = \ln(102/95) = 0.0713$
2. $\bar{r} = (0.0296 - 0.0498 - 0.0311 + 0.0713)/4 = 0.0050$; $s^2 = \frac{1}{3} \sum (r_i - \bar{r})^2 = 0.00253$; $s = 0.0503$.
Skewness: $S \approx 0.56$; Kurtosis: $K \approx 1.72$
3. $JB = \frac{4}{6}(0.56^2 + (1.72 - 3)^2/4) = 0.48$. With $\chi^2_2(0.05) = 5.99$: $0.48 < 5.99 \Rightarrow$ **do not reject** normality (sample too small!)
4. $r_{0:4} = \ln(102/100) = 0.0198 = r_1 + r_2 + r_3 + r_4 = 0.0198 \checkmark$

Exercise 2: Solution

Solution

1. $\text{VaR}_\alpha = -(\mu + z_\alpha \sigma) \times W$, with $\mu = 0.0004$, $\sigma = 0.012$, $W = \$5\text{M}$:
► $\text{VaR } 5\%: 0.01934 \times 5\text{M} = \$96,680$; $\text{VaR } 1\%: \$137,560$; $\text{VaR } 0.1\%: \$183,400$
2. $\text{ES}_{1\%} = \sigma \cdot \phi(-2.326)/0.01 = 0.03198 \Rightarrow \$159,900$
3. $P(r < -0.04) = \Phi(-3.37) \approx 0.038\%$ (1 day in $\sim 2,600$)
4. Empirically, events $> 3\sigma$ occur much more frequently $\Rightarrow$ Normal VaR underestimates risk

Exercise 3: Solution

Solution

1. With $\ell_N = 12.5$, $\ell_t = 14.8$, $k_N = 2$, $k_t = 3$, $n = 4$:
► Normal: $\text{AIC} = -21.0$; $\text{BIC} = -22.23$. Student- t : $\text{AIC} = -23.6$; $\text{BIC} = -25.44$
2. Student- t wins on both criteria ($\text{AIC}: -23.6 < -21.0$; $\text{BIC}: -25.44 < -22.23$)
3. Excess kurtosis: $\kappa = 6/(\nu - 4) = 6/(5 - 4) = 6$. Total kurtosis: $3 + 6 = 9$
4. The Student- t PDF decays as $|x|^{-(\nu+1)}$ (power law) vs. $e^{-x^2/2}$ for Normal. The slower decay allocates more probability to the tails

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Online resources and code

- ▣ **Quantlet:** <https://quantlet.com> ➤ Code platform for quantitative methods
- ▣ **Quantinar:** <https://quantinar.com> ➤ Learning platform for quantitative methods
- ▣ **GitHub SFM:** https://github.com/QuantLet/Stat_fin_markets/tree/master/Ch_02 ➤ Python code for Chapter 2

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Recent relevant events

- ▣ **Covid-19 (Mar. 2020):** S&P 500 –12% in one day; the 3rd worst event in history
- ▣ **FTX/crypto (Nov. 2022):** –65% collapse in 48h; extreme heavy tails
- ▣ **SVB (Mar. 2023):** Banking contagion; tail risk in the financial sector
- ▣ **Lesson:** Heavy-tailed models are **essential**, not optional

Key Takeaways — What to Remember from This Chapter

5 fundamental lessons

1. The Normal distribution **systematically underestimates** tail risk
2. Student- t ($\hat{\nu} \in [3, 8]$) captures heavy tails with a single extra parameter; decisively wins on AIC/BIC
3. Pareto and power laws unify tail theory: α connects Student- t , stable distributions, and EVT
4. ES is more informative than VaR — Normal underestimates ES by 30–50%
5. **The distribution choice directly determines regulatory capital:** Basel III FRTB (2023) mandates 2.5% ES

Preparation for upcoming chapters

- ▣ **Ch. 3:** Stable distributions, skew- t , GED — asymmetry and heavier tails
- ▣ **Ch. 4:** Extreme Value Theory (EVT) — generalized Pareto, thresholds

Thank You!

Questions?

Course materials are available at: <https://danpele.github.io/SFM/>



Quantlet



Quantinar